# FLEN: Leveraging Field for Scalable CTR Prediction

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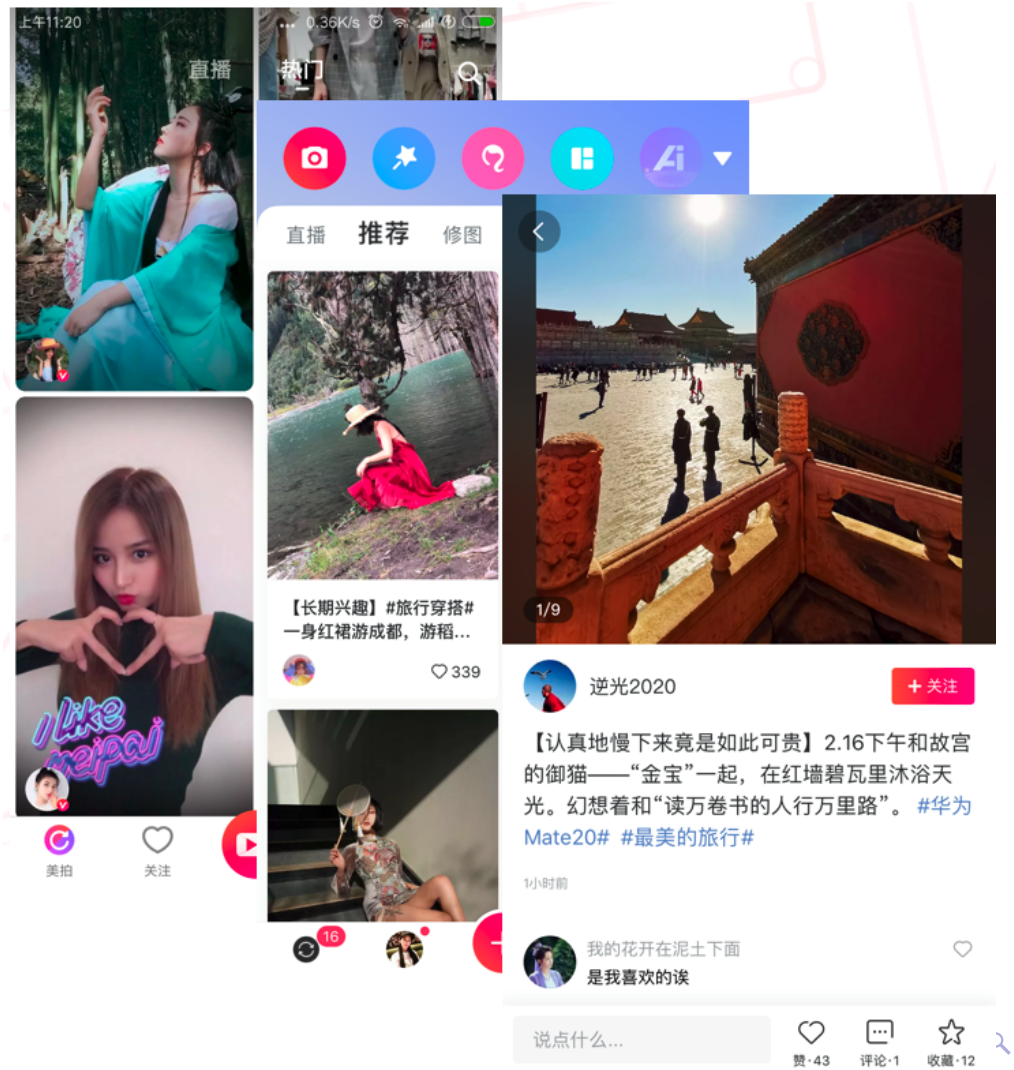
# Challenges In CTR prediction task

## Meitu app feed as an example

- Large scale
  - Over one hundred million users
  - Millions of items
- Strict latency
  - Billions of request everyday
  - Response in less then 10ms



# Existing Works and their limitation

**FFM, FwFM, xDeepFM, NFFM**

- Pros:
  - Leverage field information.
  - Achieve promising results.
- Limitations:
  - Unacceptable parameter size
  - Unacceptable computation time.

# Overall of Our Solution

- Factorized FwFM into the combination of Matrix Factorization and Factorizatoin Machines
  - captures both inter-field and intra-field feature conjunctions
  - acceptable model parameters and time complexity
- Dicefactor to prevent independent latent features from co-adapting.

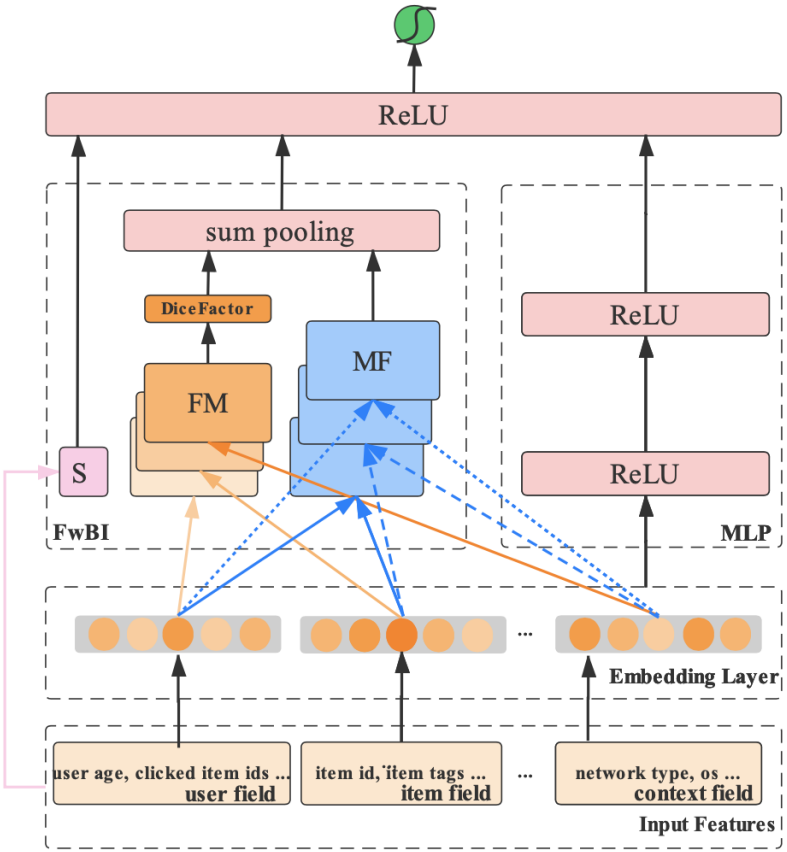



Figure: Architecture overview of FLEN.

# Feature representation

## Hierarchical field structure

$$\underbrace{\underbrace{[0,1,0,\ldots,0]}_{\textit{age field}} \ldots \underbrace{[\quad 1,0 \quad]}_{\textit{gender field}}}_{\textit{user field}} \; \underbrace{\underbrace{[0,1,0,\ldots,0]}_{\textit{item id field}} \; \underbrace{[0,1,0,1,\ldots,0]}_{\textit{item tags field}}}_{\textit{item field}}$$

## Embedding

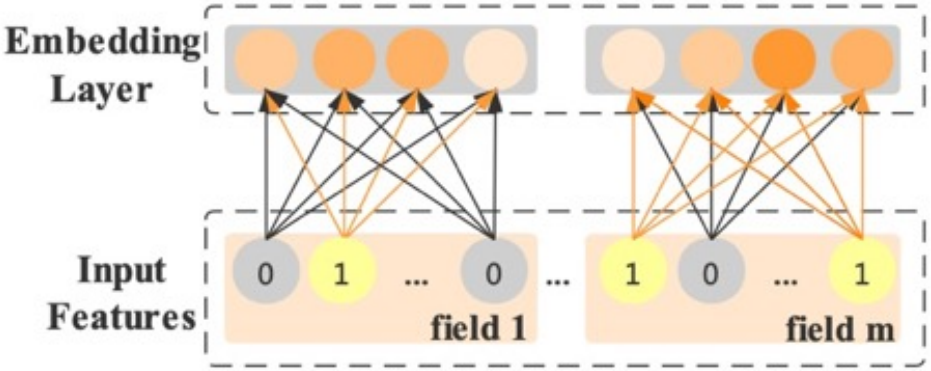



**Figure 2: Illustration of embedding layer with $K_e = 4$.**

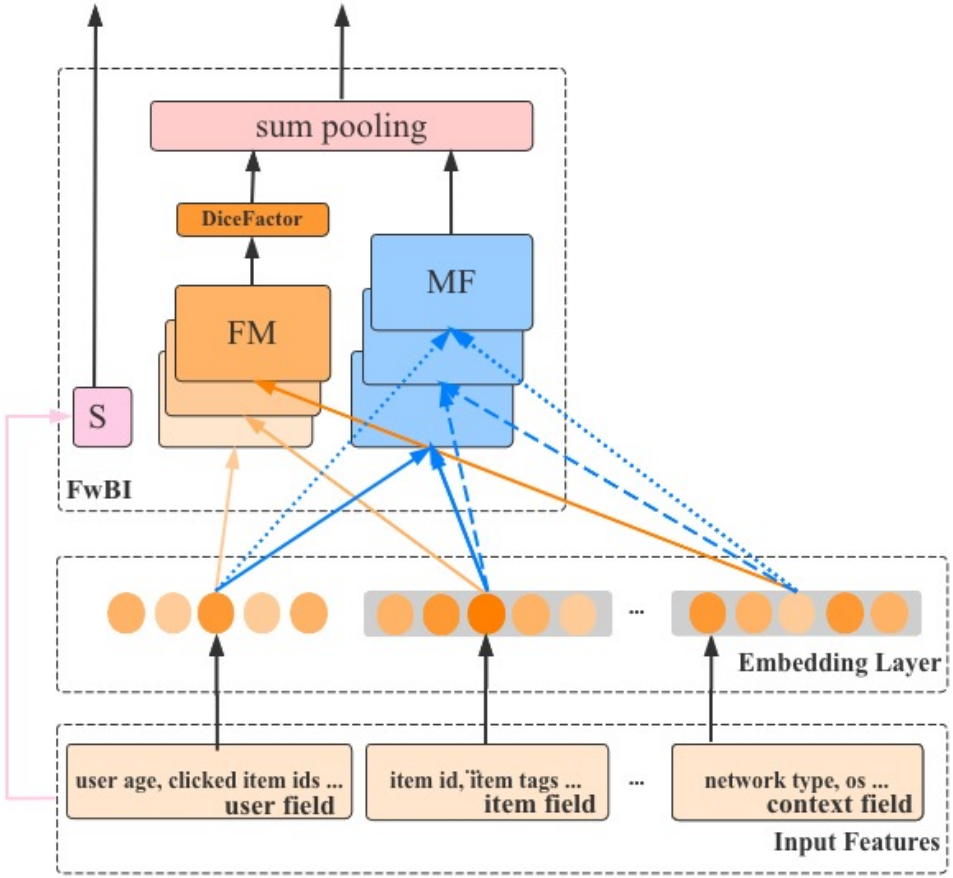
Figure: Illustration of embedding layer with $K_e = 4$.

# Field-wise Bi-Interaction Pooling(1/2)

$$\Phi_{FwBI} = \mathbf{W}_{FwBI}^T \Big[ \underbrace{w_0 + \sum_{i=1}^{\sum_n^N K_n} w[i]x[i]}_{S}$$

$$, \underbrace{\sum_{i=1}^{M} \sum_{j=i+1}^{M} \Big[ \Big( \sum_{n,F(n)=i} \mathbf{V}_i \mathbf{x}_n \Big) \Big( \sum_{n,F(n)=j} \mathbf{V}_j \mathbf{x}_n \Big) r[i][j] \Big]}_{MF}$$

$$+ \underbrace{\sum_{m}^{M} \Big[ \Big( \sum_{n,F(n)=m} \mathbf{V}_n \mathbf{x}_n \Big)^2 - \Big( \sum_{n,F(n)=m} (\mathbf{V}_n \mathbf{x}_n)^2 \Big) \Big] r[m][m] \Big]}_{FM} \quad (1)$$



# Field-wise Bi-Interaction Pooling(2/2)

$$\Phi_{FwBI} = \mathbf{W}_{FwBI}^{T}\Big[\underbrace{w_0 + \sum_{i=1}^{\sum_n^N K_n} w[i]x[i]}_{S}$$

$$,\underbrace{\sum_{i=1}^{M}\sum_{j=i+1}^{M}\Big[\Big(\sum_{n,F(n)=i}\mathbf{V}_i\mathbf{x}_n\Big)\Big(\sum_{n,F(n)=j}\mathbf{V}_j\mathbf{x}_n\Big)r[i][j]\Big]}_{MF}$$

$$+\underbrace{\sum_{m}^{M}\Big[\Big(\sum_{n,F(n)=m}\mathbf{V}_n\mathbf{x}_n\Big)^2-\Big(\sum_{n,F(n)=m}(\mathbf{V}_n\mathbf{x}_n)^2\Big)\Big]r[m][m]\Big]}_{FM} \quad (2)$$

## Model complexity

✓ *M: Number of hierarchical field number. Usually less than 4*

$$O(MK_eN + K_eM^2).$$

✓ *N: Number of categorical features*

✓ *Ke: Embedding size for each categorical feature*

# DiceFactor

- Co-training issue of FM

$$\nabla_{v_{\text{gender}}} f_{\text{FM}}(x) = v_{\text{gender}} + \ldots + v_{\text{network}} \tag{3}$$

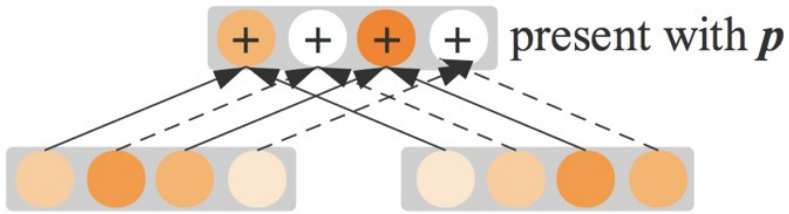
cann't decouple the gradient update process of $v_{\text{gender}}$ and $v_{\text{network}}$

- DiceFactor

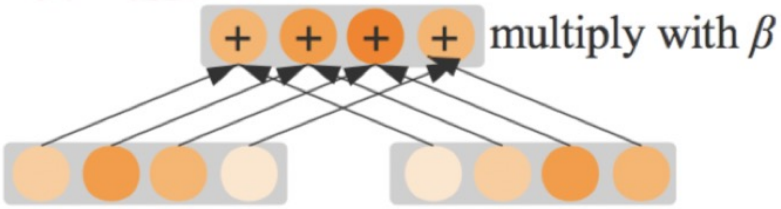
$$\Phi_{FM} = \left[ w_0 + \sum_{i=1}^{N} w[i]x[i], \sum_{i=1}^{M} \sum_{j=1 \& i \neq j}^{M} \mathbf{pe}_i \odot \mathbf{e}_j \right] \tag{4}$$

where $\mathbf{p} \in \Re^{K_e}$ and $p[i] \sim Bernoulli(\beta)$

$$\Phi_{FM} = \left[ w_0 + \sum_{i=1}^{N} w[i]x[i], \sum_{i}^{M} \sum_{j \& i \neq j}^{M} \beta \mathbf{e}_i \odot \mathbf{e}_j \right] \tag{5}$$




(a) train phase




(b) inference phase

# Datasets

Table: Statistics of evaluation data sets.

| Data | #Samples | #Fields | #Features (Sparse) |
|---|---|---|---|
| Avazu | 40,428,967 | 22 | 32,699 |
| Meitu | 1,508,149,301 | 33 | 5,476,029 |

# Datasets

Table: Example features in Meitu dataset.

| Field | Feature | Dimemsionality | Type | AverageNonzero Ids per Instance |
|---|---|---|---|---|
| User Field | gender | 2 | one-hot | 1 |
| | age | $\sim 10$ | one-hot | 1 |
| | clicked_feed_ids | $\sim 10^7$ | multi-hot | $\sim 10^2$ |
| | liked_feed_ids | $\sim 10^5$ | multi-hot | $\sim 10^2$ |
| Item Field | feed_id | $\sim 10^7$ | one-hot | 1 |
| | tags | $\sim 10^3$ | multi-hot | $\sim 10^1$ |
| | clicked_rate | 10 | one-hot | 10 |
| | liked_rate | 10 | one-hot | 10 |
| | author_id | $\sim 10^7$ | one-hot | 1 |
| | author_gender | 2 | one-hot | 1 |
| | author_age | $\sim 10$ | one-hot | 1 |
| | author_clicked_rate | 10 | one-hot | 10 |
| | author_liked_rate | 10 | one-hot | 10 |
| Context Field | network_type | $\sim 4$ | one-hot | 1 |
| | time | $\sim 10$ | one-hot | 1 |
| | brand | $\sim 15$ | one-hot | 1 |

# Competitors

We compare FLEN with 7 state-of-the-art models:

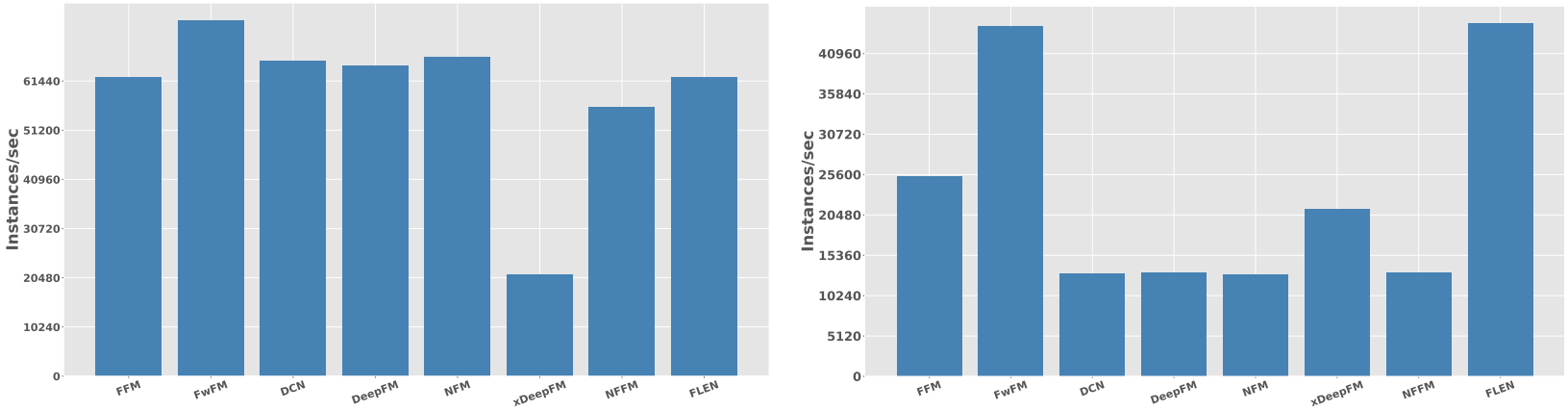
- **FFM**: It represents a feature by several separate vectors, depending on the multiplying feature field.
- **FwFM**: Explicitly aggregates over feature products. Interaction weights are assigned for each field pair.
- **DCN**: Takes the outer product of feature vectors at bit-wise level to a feed-forward neural network.
- **DeepFM**: Consists of a wide component that models factorization machine and a deep component.
- **NFM**: Stacks MLP on top of a bi-interaction pooling layer.
- **xDeepFM**: Explicitly generates features with a compressed interaction network.
- **NFFM**: Learns different feature representations for convolutional operations and product operations.

# AUC and Logloss

Table: AUC and Logloss performance of different models

| Model | Avazu | | Meitu | |
|---|---|---|---|---|
| | AUC | Logloss | AUC | Logloss |
| FFM | 0.7400 | 0.3994 | 0.6300 | 0.5625 |
| FwFM | 0.7406 | 0.3988 | 0.6306 | 0.5621 |
| DCN | 0.7421 | 0.3981 | 0.6337 | 0.5606 |
| DeepFM | 0.7438 | 0.3982 | 0.6329 | 0.5612 |
| NFM | 0.7449 | 0.3973 | 0.6359 | 0.5596 |
| xDeepFM | 0.7509 | 0.3947 | 0.6440 | 0.5576 |
| NFFM | 0.7513 | 0.3945 | 0.6443 | 0.5565 |
| FLEN | **0.7519** | **0.3944** | **0.6463** | **0.5558** |
| FLEN+D | **0.7528** | **0.3944** | **0.6475** | **0.5554** |

# TPS of different models




(a) (b)

Figure: Number of instances per second processed by different models

# Impact of Parameters

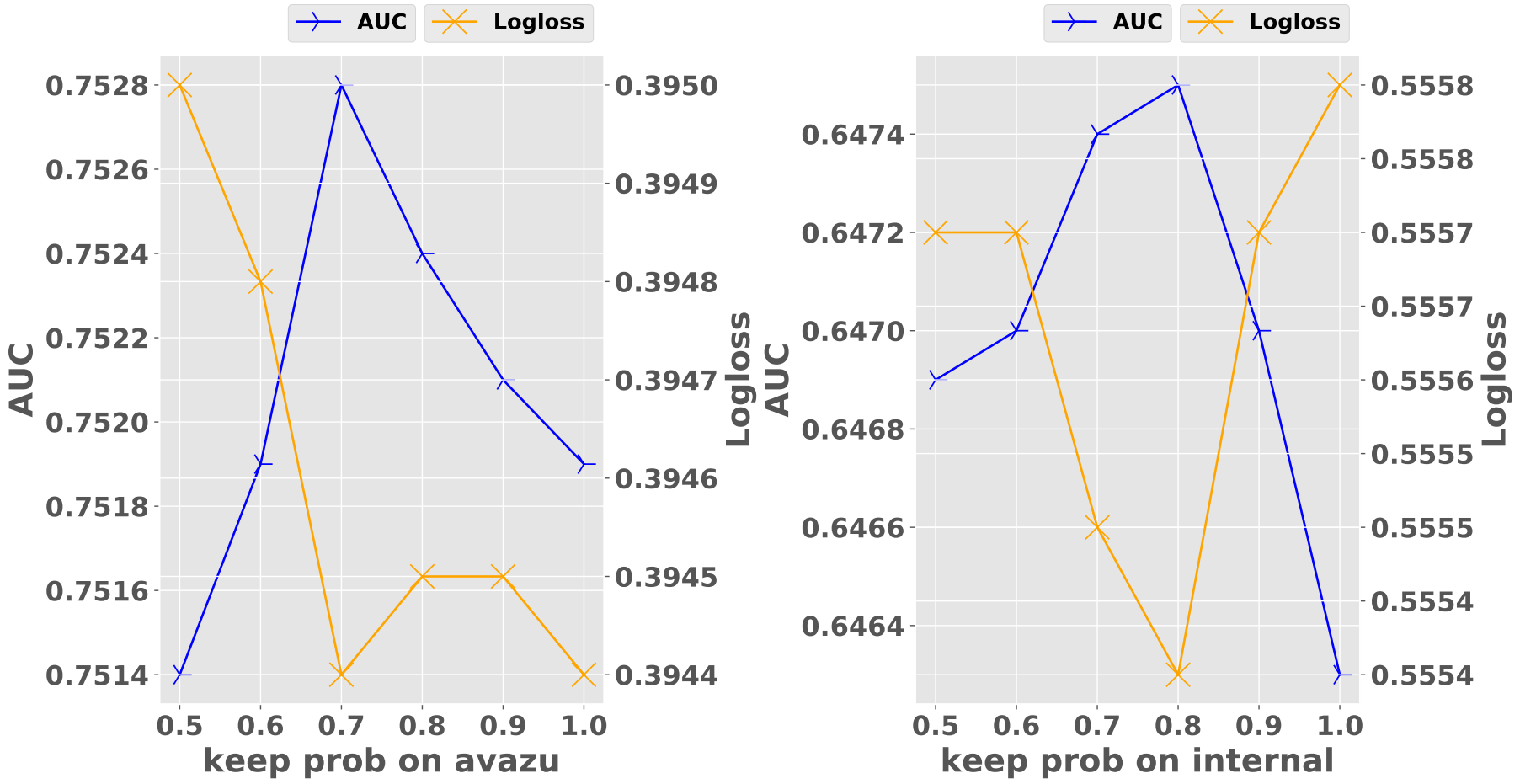



Figure: AUC and logloss with different keep probability of dicefactor on two datasets.

# Online Evaluation

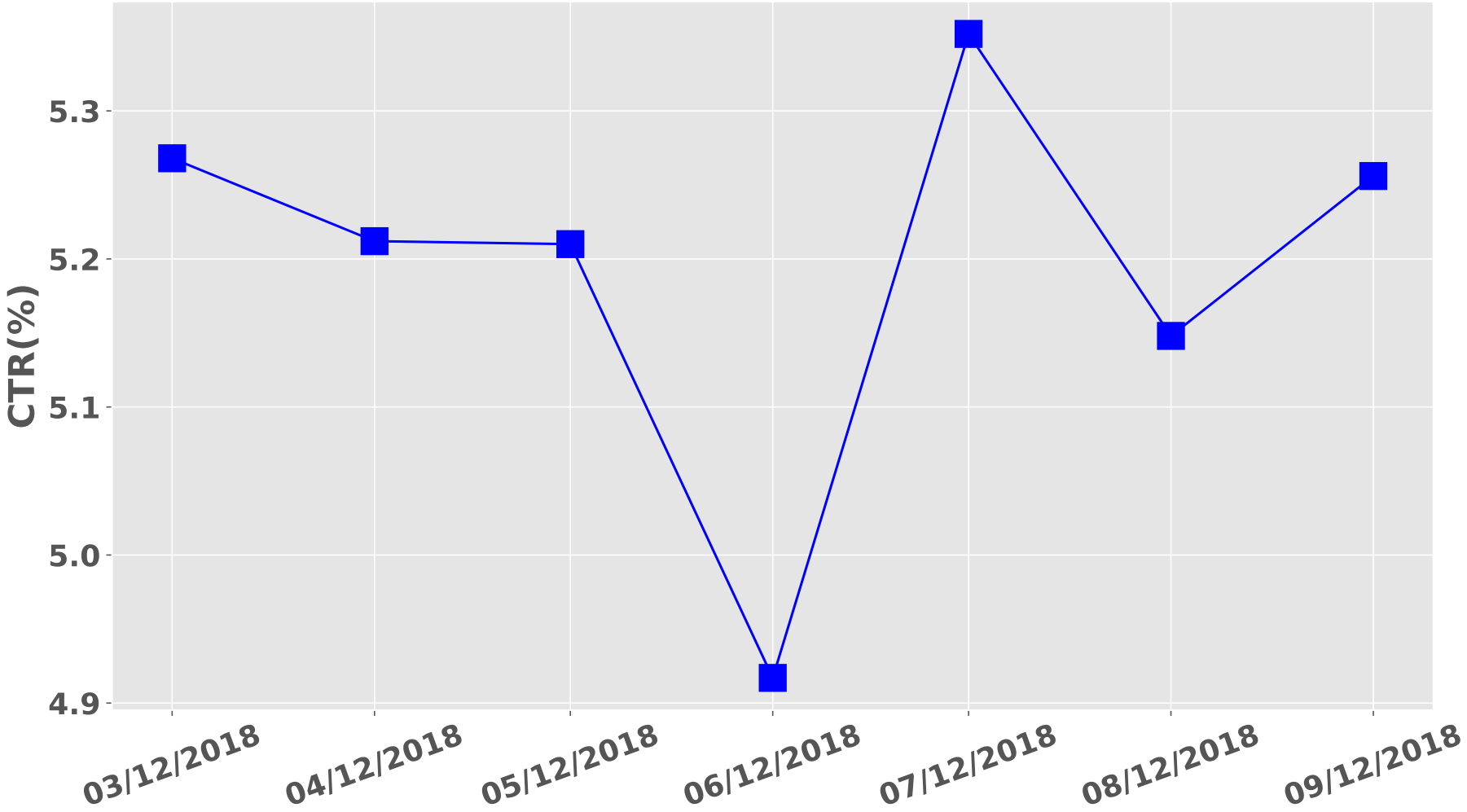



Figure: CTR increase during our online A/B testing period.

# Conclusions

- We propose FLEN model which has been deployed in the commercial recommender systems of Meitu and serves the main traffic.
- FLEN devises a field-wise bi-interaction pooling technique. It captures both inter-field and intra-field feature conjunctions with an acceptable time complexity for industrial applications.
- We develop Dicefactor: a novel dropout method to prevent independent latent features from co-adapting.

# Thank you!


- We thank the anonymous reviewers for their expert and constructive comments and suggestions.
- We thank Mr. Runquan Xie for his helpful comments and discussions.
- We thank Mr. Haoxuan Huang for his discussions and help in the extensive experiments at Meitu.

- Codes and slides are available at: `https://github.com/aimetrics/jarvis`
- If you have any questions, please feel free to contact us.